
A Confirmed Mechanism With an Unreachable Operating Point

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Abstract

We report the first mechanism in this campaign whose discriminating prediction was registered before the discriminating data existed and then held. A profile attributes the training step’s low utilisation ($\approx 19\%$ MFU, launch-bound) to $\approx 194k$ tiny fill launches in the hashed n -gram value-embedding path. We proposed a cause — autograd’s dense embedding backward allocates and zero-fills a gradient the size of the entire table while a step touches $\sim 1e-4$ of its rows — and derived a prediction that separates it from the obvious rival: fill-flood says the cost of the dense path scales with TABLE ROWS, so a sparse-gradient replacement should look better as tables grow; a memory-bandwidth account says it scales with TOUCHED rows, so the comparison should be flat in table size. A first experiment (E6) tested the sparse replacement at the frame’s default multiplier and failed in the direction opposite to its own prediction: $+13.4\%$ step time, killed at its preregistered trap-check. A second (E7), a 2×2 over table multiplier and sparse gradients, then confirmed the interaction: the sparse penalty falls from $+12.95\%$ to $+3.22\%$ as tables grow $4\times$. The mechanism is supported and the intervention is dead, because the table size at which sparse gradients would pay ($\approx 342\times$) lies far past the quality knee: $256\times$ already costs $+0.0109$ val_bpb, $4.2\times$ the adoption gate. We argue this pairing — a true mechanism with an unaffordable operating point — is a distinct and under-reported outcome class, and that separating “the mechanism is right” from “the change is worth making” is what let us extract a positive result from two failed experiments.

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1. Why This Paper Is Not a Third Null

Papers 026 and 027 recorded five preregistered interventions landing inside $\pm 1.3e-3$ of zero against a gate of $2.6e-3$. The natural next move — run a sixth — has poor expected information. What changed here is that we stopped testing interventions and started testing a mechanism: a proposed cause for an observed phenomenon, carrying a prediction that some rival account forbids.

The distinction matters operationally. An intervention test answers “does this help?” and a null answers almost nothing about why. A mechanism test answers “is this the reason?” and either outcome is informative. Both E6 and E7 were, as interventions, failures. As a mechanism test the pair is a success.

2. The Observed Phenomenon

Profiling the reference step (evd_obs_ngram_kernel_profile) shows the step is dominated not by attention ($\approx 5\%$) or cross-entropy ($\approx 1.6\%$) but by a long tail of tiny kernels, including $\approx 194k$ int/fill launches concentrated in the n -gram value-embedding construction. Measured MFU is $\approx 19\%$ while paper 026 measured the block GEMMs themselves at $\approx 90\%$. The loss is outside the matmuls, and the step is launch-bound, not compute-bound.

3. The Proposed Cause and Its Falsifier

The hashed n -gram value embeddings hold $\approx 1.61e9$ parameters over 14 registered hash sites — 95% of the model’s real parameter count, a fact absent from the reported 94M figure. A single step gathers at most $B \cdot T \approx 1.5e5$ distinct rows per site, a $\sim 1e-4$ fraction. Autograd’s dense embedding backward nevertheless allocates a gradient the full size of the table and must zero it before scatter-accumulation. Emitted as many small tiled fills rather than one memset, that zeroing converts table size into launch count — which is what the profile shows.

The registered discriminator. Let $P(m)$ be the step-time penalty of replacing the dense backward with

sparse capture at table multiplier m , and let O be the fixed overhead sparse capture adds (chiefly the custom_op boundary blocking inductor’s epilogue fusion). Fill-flood predicts the removed work grows with table rows:

$$P(m) = O - F \cdot (m/64),$$

strictly decreasing in m . The bandwidth account predicts P independent of m . This was written into the mechanism record before E6 returned any data.

4. E6: Failing in the Opposite Direction

E6 enabled sparse capture at the default multiplier, predicting -9.3% step time and val_bpb -0.0058 via the token law. Measured over 5 clean pairs: 1779 vs 2016 mean steps, $+13.37\%$ step time, 5/5 pairs against, every treatment arm below every control arm. The round was killed at its preregistered batch-1 trap-check rather than run to $n=10$.

Two things are worth recording. First, the falsification record had named the expected failure mode in advance — “the custom_op boundary blocks inductor epilogue fusion and the lost fusion costs more than the removed fills” — and that is what occurred; this is the seventh proposal killed by the campaign’s compiled-baseline rule. Second, the analysis rule was fixed before launch: because the treatment’s own mechanism is throughput, RAW was preregistered as the decision metric. Applying the token-law control variate here would have subtracted the hypothesis, converting “13% slower” into “intrinsic $+0.0009$, essentially null” — a defensible-sounding sentence that would have been false by construction.

An unplanned check of the token law. It predicts $+0.00753$ val_bpb from the lost tokens alone; the observed RAW delta was $+0.00841$. Agreement to 12% at $\approx 3\times$ the throughput perturbation of any prior test, leaving an intrinsic residual of $+0.0009$.

5. E7: The Interaction

A 2×2 over $m \in \{64, 256\}$ and sparse $\in \{0, 1\}$, 4 seeds per cell, 16 arms.

cell	val_bpb	step ms	med MFU
$m=64$, dense	0.933346	149.8	19.41
$m=64$, sparse	0.941059	169.2	17.18
$m=256$, dense	0.944230	186.2	15.62
$m=256$, sparse	0.946569	192.2	15.13

Penalty at $m=64$: $+19.4$ ms ($+12.95\%$). At $m=256$: $+6.0$ ms ($+3.22\%$). The penalty falls $3.2\times$ while tables grow $4\times$. The prediction registered before E6 is

confirmed; the bandwidth account, which requires flatness, is contradicted.

Solving $P(m) = O - F(m/64)$ on the two points gives $F = 4.47$ ms of fill cost at $m=64$ and $O = 23.87$ ms of custom_op overhead, hence break-even near $m \approx 342$. We flag this explicitly as an interpolation, not a test — two points determine two parameters exactly — and register it as a prediction for a third multiplier rather than a result.

A methodological note that changed a verdict. Our contention screen uses a ≥ 1950 -step floor, calibrated at $m=64$. The $m=256$ arms legitimately complete ≈ 1620 steps because the step is 25% slower; an absolute floor would have quarantined an entire healthy cell. Median step time is operating-point-independent and contention-robust, and screening on it also exonerated arm s45_m64_sg0, which the step floor would have flagged (1908 steps) despite a step time of 150.2 ms, squarely in family. The step-count screen is a proxy that silently assumes a fixed operating point.

6. Why the Direction Is Nevertheless Closed

The regime in which the fill flood dominates is a regime this frame cannot afford. At fixed sparsity setting, $m=256$ costs $+0.010883$ val_bpb against $m=64$ — $4.2\times$ the gate. Break-even for sparse gradients sits above that, at $m \approx 342$. There is no table size that is both large enough for sparse gradients to pay and small enough for the model to be good.

This is a specific and, we think, under-named outcome: the mechanism is real, its prediction survived a prospective discriminating test, and the intervention derived from it is still worthless here. Recording only “sparse gradients failed” would have discarded a confirmed causal account of the campaign’s dominant throughput bottleneck. Recording only “fill-flood confirmed” would have implied an adoption that does not exist. The registry now carries them as separate records: mechanism status=active, hypothesis status=rejected.

7. Next Experiments

Rated on novelty / provenance / validity / impact / reliability / feasibility / falsifiability, 1–5.

1. TN-gram: CP factors shared across n -gram orders (5/5/4/5/3/4/5) — running as E8. One shared factor table replaces the 14 order- and layer-specific tables, with a learned per-site absorption vector specialising each use. Why it should work: clm_tngram_cp_shared_factors_mechanism

is the only claim in our 123-claim corpus whose recorded scope is literally `short_budget_transformer_lm_pretraining` — no downward extrapolation is required. It attacks the subsystem holding 95% of parameters and, per this paper, the throughput bottleneck: 14 dense zero-fills collapse to 1. Smoke measurement gives 143.1ms vs 148.9ms (−3.9%) and 44 GB vs 55 GB. Design: three arms, because two would confound sharing with capacity. A = baseline; B = shared factors ($\approx 14\times$ fewer n -gram parameters); C = independent tables shrunk to B’s parameter count. B vs C isolates sharing; A vs C measures capacity alone. Prior against it: E7 shows the capacity-quality curve is steep, so B pays a capacity cost that sharing must overcome.

2. A third multiplier to test the $P(m)$ fit (2/5/5/2/5/5/5). Run sparse vs dense at $m=128$; the fit predicts +14.9ms. Cheap, and converts an interpolation into a test. Low impact — it cannot reopen the direction — but it is the honest completion of this paper’s central claim.

3. Attack O rather than F (4/3/4/4/3/3/4). The 23.87ms `custom_op` overhead is now measured, not inferred, and it is the single largest identified non-GEMM cost in the step. If sparse capture were expressible without an autograd-function boundary — e.g. as a compiled scatter that inductor can fuse — break-even moves from $m \approx 342$ toward the usable range. Risk: six proposals have already died to this exact boundary, which is evidence the boundary is hard to avoid, not that nobody tried.

4. Retire the absolute step-count screen (1/5/5/3/5/5/4). Replace the ≥ 1950 -step floor with a median-step-time band computed per operating point. This paper shows the floor mis-screens whenever an arm legitimately changes step time, which is exactly the case for every throughput lever we intend to test.

8. Conclusion

Two failed interventions produced one confirmed mechanism. The fill-flood account of this model’s launch-bound step made a prediction that a rival account forbade, the prediction was registered before the data existed, and it held: the sparse-gradient penalty fell $3.2\times$ as tables grew $4\times$. The same experiment closed the direction, because the multiplier at which the mechanism pays costs 4.2 gates of quality. We record the mechanism as active and the intervention as rejected, and we take the general lesson to be that a

campaign stuck on nulls should test causes rather than changes — a null intervention teaches almost nothing, while a mechanism test cannot fail to.